

Martina Legris

Assistant professor

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PROFILE

I am a plant biologist interested in molecular signaling of photoreceptors and developmental plasticity in response to light and temperature signals. I am passionate about doing research and communicating science, both at university courses as well as to broader audiences.

PAST POSITIONS

Postdoctoral researcher at Christian Fankhauser's lab. Center for Integrative Genomics, Faculty of Biology and Medicine, University of Lausanne, Switzerland. August 2017 – February 2024.

EDUCATION

Ph. D. in Biology. April 2012 - May 2017

School of Exact and Natural Sciences, University of Buenos Aires, Argentina and Leloir Institute, Buenos Aires, Argentina.

Thesis topic: Molecular basis of light and temperature signals crosstalk in *Arabidopsis thaliana*.
Supervisor: Jorge J. Casal.

B. Sc. / M. Sc. in Biology. September 2006 - March 2012

School of Exact and Natural Sciences, University of Buenos Aires, Argentina and Leloir Institute, Buenos Aires, Argentina.

Thesis topic: Molecular and Physiological characterization of *pICln* function in *Arabidopsis thaliana*.
Supervisor: Marcelo J. Yanovsky

PUBLICATIONS

* Equal contributions. § Corresponding author.

Iglesias MJ[§], Costigliolo-Rojas C, Bianchimano L, **Legris M**, Schön J, Gergoff-Grozeff GE, Bartoli CG, Blázquez MA, Alabadí D, Zurbriggen MD and Casal JJ[§]. 2024. Shade-induced ROS/NO reinforce COP1-mediated diffuse cell growth. **Proceedings of the National Academy of Sciences** 121(42), p.e2320187121

Nawkar GM*, **Legris M***, Goyal A, Schmid-Siebert E, Fleury J, Mucciolo A, De Bellis D, Trevisan M, Schueler A and Fankhauser C[§]. 2023. Air channels create a directional light signal to regulate hypocotyl phototropism. **Science**, 382(6673), pp.935-940.

Legris M[§], 2023. Light and temperature regulation of leaf morphogenesis in *Arabidopsis*. **New Phytologist**, 240(6), pp.2191-2196.

Lopez Vazquez A, Allenbach Petrolati L, **Legris M**, Dessimoz C, Lampugnani ER, Glover N, Fankhauser C[§]. 2023. Protein S-acylation controls the subcellular localization and biological activity of PHYTOCHROME KINASE SUBSTRATE. **Plant Cell**. doi.org/10.1093/plcell/koad096

Mateos JL, Sanchez SE, **Legris M**, Esteve-Bruna D, Torchio JC, Petrillo E, Goretti D, Blanco-Touriñán N, Seymour DK, Schmid M, Weigel D, Alabadí D, Yanovsky MJ. 2022. PICLN modulates alternative splicing and light/temperature responses in plants. **Plant physiology**, kiac527, <https://doi.org/10.1093/plphys/kiac527>

Nieto C*, Catalán P*, Luengo LM, **Legrís M**, López-Salmerón V, Davière JM, Casal JJ, Ares S[§], Prat S[§]. 2022. COP1 dynamics integrate conflicting seasonal light and thermal cues in the control of Arabidopsis elongation. **Science Advances**, 8(33), p.eabp8412.

Legrís M, Szarzynska-Erden BM, Trevisan M, Allenbach-Petrolati L, Fankhauser C[§]. 2021. Phototropin-mediated perception of light direction in leaves regulates blade flattening. **Plant physiology**, 187(3):1235-1249.

Boccaccini A, **Legrís M**, Krahmer J, Allenbach-Petrolati L, Goyal A, Galvan-Ampudia C, Vernoux T, Karayekov E, Casal JJ and Fankhauser C[§]. 2020. Low Blue Light Enhances Phototropism by Releasing Cryptochrome1-Mediated Inhibition of PIF4 Expression. **Plant physiology**, 183(4), pp.1780-1793.

Legrís M[§], Boccaccini A. 2020. Stem phototropism towards blue and ultraviolet light. **Physiologia Plantarum**. doi:10.1111/ppl.13098.

Blanco-Touriñán N*, **Legrís M***, Minguet EG*, Costigliolo-Rojas C*, Nohales MA, Iniesto E, García-León M, Pacín M, Heucken N, Blomeier T, Locascio A, Černý M, Esteve-Bruna D, Díez-Díaz M, Brzobohatý B, Frerigmann H, Zurbriggen MD, Kay SA, Rubio V, Blázquez MA, Casal JJ[§], Alabadi D[§]. 2020. COP1 destabilizes DELLA proteins in Arabidopsis. **Proceedings of the National Academy of Sciences**. DOI: 10.1073/pnas.1907969117

Legrís M, Ince YÇ and Fankhauser C[§]. 2019. Molecular mechanisms underlying phytochrome-controlled morphogenesis in plants. **Nature communications**, 10(1), pp.1-15.

Sellaro R, Smith RW, **Legrís M**, Fleck C[§] and Casal JJ[§]. 2019. Phytochrome B dynamics departs from photoequilibrium in the field. **Plant, cell & environment**, 42(2), pp.606-617.

Arico D, **Legrís M**, Castro L, Garcia CF, Laino A, Casal JJ[§] and Mazzella MA[§]. 2019. Neighbour signals perceived by phytochrome B increase thermotolerance in Arabidopsis. **Plant, cell & environment**, 42(9), pp.2554-2566.

Cagnola JI, Cerdan PD, Pacin M, Andrade A, Rodriguez V, Zurbriggen MD, **Legrís M**, Buchovsky S, Carrillo N, Chory J, Blazquez MA, Alabadi D, Casal JJ[§]. 2018. Long-day photoperiod enhances jasmonic acid-related plant defense. **Plant physiology**, 178(1), pp.163-173.

Pucciariello O, **Legrís M***, Rojas CC*, Iglesias MJ*, Hernando CE, Dezar C, Vazquez M, Yanovsky MJ, Finlayson SA, Prat S and Casal JJ[§]. 2018. Rewiring of auxin signaling under persistent shade. **Proceedings of the National Academy of Sciences**, 115(21), pp.5612-5617.2017

Legrís M*, Nieto C*, Sellaro R, Prat S, Casal JJ[§]. 2017. Perception and signalling of light and temperature cues in plants. **Plant Journal**. 90(4):683-697.

Legrís M, Klose C*, Burgie ES*, Costigliolo C*, Neme M, Hiltbrunner A, Wigge PA, Schäfer E**, Vierstra R**, Casal JJ[§]. 2016. Phytochrome B integrates light and temperature signals in Arabidopsis. **Science**. 354(6314):897-900.

Pacín M, Semmoloni M, **Legrís M**, Finlayson SA and Casal JJ[§]. 2016. Convergence of CONSTITUTIVE PHOTOMORPHOGENESIS 1 and PHYTOCHROME INTERACTING FACTOR signalling during shade avoidance. **New Phytologist**. 211(3):967-79.

Trupkin SA*, **Legrís M***, Buchovsky AS, Tolava Rivero MB, Casal JJ[§]. 2014. Phytochrome B nuclear bodies respond to the low red / far-red ratio and to the reduced irradiance of canopy shade in Arabidopsis. **Plant Physiology**. 165(4):1698-1708.

Pacín M*, **Legrís M***, Casal JJ[§]. 2014. Rapid Decline in Nuclear COSTITUTIVE PHOTOMORPHOGENESIS1 Abundance Anticipates the Stabilization of Its Target ELONGATED HYPOCOTYL5 in the Light. **Plant Physiology**. 164(3):1134-8.

Karayekov E*, Sellaro R*, **Legrís M**, Yanovsky MJ, Casal JJ[§]. 2013. Heat shock-induced fluctuations in clock and light signaling enhance phytochrome B-mediated Arabidopsis deetiolation. **Plant Cell**. 25(8):2892-906.

Pacín M, **Legris M**, Casal JJ[§]. 2013. COP1 re-accumulates in the nucleus under shade. **Plant Journal**. 75(4):631-41.

CONFERENCE PRESENTATIONS

- Invited speaker.** IV Young Investigators Symposium. IBMCP (UPV-CSIC). September 20, 2024. Valencia, Spain. *"Perception of directional light cues in Arabidopsis"*.
- Talk.** International Symposium on Plant Photobiology. January 9 – 13, 2024. Bhubaneswar, India. *"Establishment of a light gradient for phototropism"*.
- Talk.** Swiss PLANT meeting. June 13 – 15, 2022. Meiringen, Switzerland. *"Phototropins perceive light direction in the leaf to regulate blade flattening"*.
- Invited speaker.** International Symposium on Plant Photobiology. June 3 - 8, 2019. Barcelona, Spain. *"Mechanisms underlying phototropin regulation of leaf flattening"*.
- Talk.** International symposium Auxins and Cytokinins in Plant Development. July 1 - 5, 2018. Prague, Czech Republic. *"Light control of leaf flattening"*.
- Talk.** Sainsbury Laboratory Symposium. The coordination of development. September 19 - 21, 2018. Cambridge, UK. *"How to grow a flat leaf?"*.
- Talk.** Young Researchers Symposium in Plant Photobiology. 3-6 October, 2018. Cologne, Germany. *"How to grow a flat leaf?"*
- Talk.** Annual Meeting of the Brazilian Society of Biochemistry and Molecular Biology. July 27 - 30, 2017. Sao Paulo, Brazil. *"Phytochrome B Integrates Light and Temperature Signals in Arabidopsis"*.
- Talk.** Annual Meeting of the Argentine Society for Biochemistry and Molecular Biology. November 7 - 10, 2016 Córdoba, Argentina. *"Integration of light and temperature cues in plant development"*.

GRANTS AND SCHOLARSHIPS

- Fondation Mercier pour la Science.** "Temperature regulation of leaf initiation in Arabidopsis thaliana". Two-year grant to hire a postdoctoral researcher and start a new research line in the lab. Awarded February 2026.
- SNSF R'equip (co-applicant).** 316030_229682 "Imaging system for multi-spectral sensitive and life-time separated fluorescence detection in complex biological samples". Grant to buy a confocal microscope for the Institute of Biology, University of Neuchâtel. Awarded in 2024.
- SNSF Starting Grant.** TMSGI3_218178. "Environmental control of shoot architecture in Arabidopsis". Five-year grant to start my independent research group as an Assistant Professor together with a PhD student and a postdoctoral researcher. Awarded in 2023.
- Tremplin grant. University of Lausanne.** One-year grant to hire technicians to support my research. Awarded in 2022.
- Marie Skłodowska-Curie Actions Individual fellowship.** Flat_Leaf 796443. Two-year postdoctoral fellowship and grant. Awarded in 2018.
- Human Frontiers Long term fellowship.** LT000829/2018-L. Three-year postdoctoral fellowship and grant. Awarded in 2018.
- EMBO Long-term fellowship.** ALTF 46-2017. Two-year postdoctoral fellowship. Awarded in 2017.
- Beca doctoral CONICET.** National Scientific and Technical Research Council of Argentina. Five-year Doctoral Fellowship. Awarded in 2011.

AWARDS

2023 New Phytologist Tansley Medal for Excellence in plant science. The New Phytologist Foundation, UK. 2023.

FBM award “Young Researcher in Basic Sciences”. University of Lausanne, Switzerland. 2022.

Finalist in 21° **Premio Joven Talento en Ciencias de la Vida**. GE Healthcare. Sao Paulo, Brasil. 2017.

Best talk in Plant Biology. 52nd annual SAIB meeting. Argentina. 2016.

Travel award and **Best project award** in course “Processing and Analysis of Fluorescence Microscopy Images”. ICGEB - Institut Pasteur. Uruguay. 2016.

TEACHING EXPERIENCE

Professor. Plant Ecophysiology. Master 1st year. University of Neuchâtel, Switzerland. 2026.

Professor. Des molécules aux cellules. Bachelor 1st year. University of Neuchâtel, Switzerland. 2025.

Assistant teacher. Advanced Molecular, Cellular and Developmental Biology. University of Lausanne, Switzerland. March 2023.

Invited teacher. EPS summer school. Utrecht University, The Netherlands. August 2022.

Assistant teacher. Molecular biology. University of Lausanne, Switzerland. 2022 - 2024.

Organizer of the CUSO workshop “Plant Development – a Combined on-Site and Virtual Workshop”. University of Lausanne, Switzerland. September 3, 2021.

Assistant teacher. Experimental design. University of Lausanne, Switzerland. Three semesters, 2018 - 2021.

Assistant teacher. Microscopy workshop. University of Lausanne, Switzerland. One semester, 2017.

Invited teacher. Postgraduate course “Introduction to confocal microscopy”. Universidad del Litoral, Argentina. March 2017.

Assistant teacher. Plant Physiology. School of Agriculture, University of Buenos Aires. Six semesters, 2014 - 2017.

Teaching assistant. Introduction to Botany. School of Exact and Natural Sciences, University of Buenos Aires. Four semesters 2010 - 2012.

Teaching assistant. Graduate course “Plant Biotechnology”. School of Exact and Natural Sciences, University of Buenos Aires. October 2010.

MENTORING

Supervision of **Postdoctoral researcher** Melisa Luquet. Since June 2026.

Supervision of **Postdoctoral researcher** Carolina Huercano-Rubens. Since February 2026.

Supervision of undergraduate **student assistant** Anaëlle Kämpfen. September 2025-June 2026.

Supervision of **PhD student** Cindy Chen. University of Neuchâtel. Since September 2024.

Supervision of undergraduate **student assistant** Tara Minchin. University of Lausanne. April 2022 – August 2024.

Supervision of undergraduate **student assistant** Vincent Frasnetti. University of Lausanne. March - September 2022.

Co-supervision of **Master student** Jérôme Benedetti. University of Lausanne. October - December 2022.

Supervision of undergraduate **student assistant** Cyann Winkler. University of Lausanne. Since September 2022.

Co-supervision of **Master student** Marion Brechet. University of Lausanne. October - December 2019.

Co-supervision of **Master student** Laetitia Tcheng. University of Lausanne and ENS Paris. February - July 2019.

Supervision of **Summer student** Marion Brechet. University of Lausanne. July - August 2018.

Supervision of **Summer student** Antoine Pochon. University of Lausanne. May - July 2018.

Co-supervision of **Master student** Luciano Miguel Medrano Moreno. University of Buenos Aires. 2013 - 2016.

PARTICIPATION IN THESIS COMMITTEES

Marius Malai. Université de Neuchâtel. Supervisor: Joop Vermeer. Since March 2025.

Lisa Oskam. Utrecht University. Supervisor: Ronald Pierik. February 2025.

Luis Miguel Luengo. Universidad Autónoma de Madrid. Supervisor: Salomé Prat. September 2024.

PEER REVIEW

I participate regularly in the peer-review process for journals such as Current Biology, Plant Physiology, Plant Journal, New Phytologist, Molecular Plant, among others.

CONFERENCE ORGANIZATION

Co-organizer of the international EMBO workshop on Plant Photobiology ISPP 2026. Les Diablerets, Switzerland. September 2026.

Co-organizer of SwissPLANT symposium. Les Diablerets, Switzerland. January 2026.

SCIENCE OUTREACH

Speaker in "Pint of Science Neuchatel". Neuchatel, Switzerland. May 2026.

Organizer of an online activity in the event "Mystères de l'UNIL" organized by the University of Lausanne, Switzerland. June 2021.

Organizer of a stand in the event "Science is Wonderful!" organized by the European commission in Brussels, Belgium. September 2019.

General Organizer of "The Night of Museums", a local event where the Leloir Institute takes part offering science outreach activities. November 2016.

Member of the Organizing Board of the "The Night of Museums", a local event where the Leloir Institute takes part offering science outreach activities. Three years, 2014 - 2016.

General coordinator of the Plant Physiology Stand in the Biological Sciences Week at the School of Exact and Natural Sciences, University of Buenos Aires. 2015.

Organizer of the Plants Stand in the Biological Sciences Week at the School of Exact and Natural Sciences, University of Buenos Aires. Four years, 2011 - 2014.

LANGUAGE SKILLS

Spanish: Native.

English: Fluent. European Framework of Reference Level C1.

French: Fluent. European Framework of Reference Level B2.

Portuguese: Elementary. European Framework of Reference Level A2.

German: Beginner. European Framework of Reference Level A1.2.